Quant II

Lab 5

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2026-03-05

Today's Agenda

- IV in practice
- Weak Instruments
- Characterizing compilers
- Constructed IV - Shift-share

IV Basics

- We worry that treatment (D) and outcome (Y) are confounded by unobservables
- But suppose we can measure an exogenous **instrument** (Z) that affects Y only via the treatment

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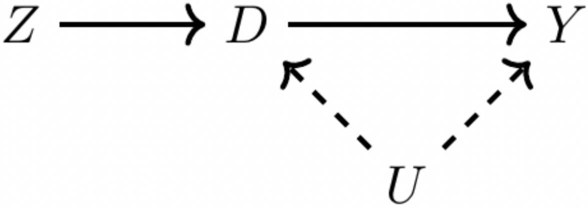

IV Basics

- Basic estimation (IV as ratio): $\frac{\text{effect of Z on Y}}{\text{effect of Z on D}}$
- 2SLS estimation:
  - Same idea, but residualize w.r.t. controls (FWL)
  - First, regress the treatment on the instruments + controls to get $\hat{D}$
  - Second, regress the outcome on the $\hat{D}$
  - This yields $\frac{Cov(\tilde{Z},\tilde{Y})}{Cov(\tilde{Z},\tilde{D})}$

2SLS

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```
s1 <- feols(education ~ nearcollege + ethnicity + smsa + south + age, sr)
sr$educ_inst <- predict(s1)
s2 <- feols(logWage ~ educ_inst + ethnicity + smsa + south + age, sr)
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```

```
                           model 1          model 2
Dependent Var.:          education          logWage

nearcollegeyes  0.3380*** (0.1072)
educ_inst                          0.0926* (0.0487)
_______________ __________________ ________________
S.E. type                      IID              IID
Observations                 3,010            3,010
---
Signif. codes: 0 '***' 0.01 '**' 0.05 '*' 0.1 ' ' 1
```

ivreg fixest estimatr

```r
# ivreg
# `y ~ x1 + x2 + ... | d | z`
pacman::p_load(ivreg, sandwich, lmtest)
m.ivreg <- ivreg(
  logWage ~ ethnicity + smsa + south + age |
    education | nearcollege,
  data = sr
)
```

ivreg fixest estimatr

```r
# fixest
# `fixest` syntax: `y ~ x1 + x2 + ... | d ~ z`
m.fixest <- feols(
  logWage ~ ethnicity + smsa + south + age |
    education ~ nearcollege,
  vcov = "hc1",
  data = sr
)
```

ivreg fixest estimatr

```r
# estimatr
# `estimatr` syntax: `y ~ x1 + x2 + ... + d | x1 + x2 + ... +
pacman::p_load(estimatr)
m.estimatr <- iv_robust(
  logWage ~ ethnicity + smsa + south + age + education |
    ethnicity + smsa + south + age + nearcollege,
  se_type = "HC1",
  data = sr,
)
```

ivreg fixest estimatr

```
models <- list(m.ivreg, m.fixest, m.estimatr)
names(models) <- c("m.ivreg", "m.fixest", "m.estimatr")
vcovs <- list(
  vcovHC(m.ivreg, type = "HC1"),
  vcov(m.fixest), vcov(m.estimatr)
)
modelsummary(models, vcov = vcovs,
             keep = c("educ"),
             statistic = "std.error",
             stars = c("***"=0.01, "**"=0.05, "*"=0.1),
             gof_map = "nobs", output = "kableExtra")
```

	m.ivreg	m.fixest	m.estimatr
education	0.093*		0.093*
	(0.050)		(0.050)
fit_education		0.093*	
		(0.050)	
Num.Obs.	3010	3010	3010

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Weak instruments

- Weak instruments: small correlation of the instrument with the endogenous treatment
- Exacerbate bias and make inference unreliable

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- Exacerbate bias and make inference unreliable
- Old advice: F-statistic of the first-stage regression > 10
- New advice: > 104.7

Weak instruments

- Example using the schooling/wages data:

```
fitstat(m.fixest, "ivf")
```

```
F-test (1st stage), education: stat = 9.9461, p = 0.001628, or
```

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- F 10 based on assumption of homoskedastic errors, hardly satisfied (Andrews, Stock & Sun 2019)
- Solutions:
  - Robust F statistic. (Andrews, Stock & Sun 2019)
  - Effective first-stage F-statistic (Montiel Olea & Pflueger 2013)

Lal et al (2024) `ivDiag`

Political Analysis

How Much Should We Trust Instrumental Variable Estimates in Political Science? Practical Advice Based on 67 Replicated Studies

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  - ③ 2SLS estimates are inflated relative to OLS because of weak instruments
- Develop `ivDiag` package (Tutorials)

ivDiag and Robust/Effective F-stat

```
pacman::p_load(ivDiag)
ivd <- ivDiag(
  data = sr,
  Y = "logWage",
  D = "education",
  Z = "nearcollege_num",
  controls = c("ethnicity", "smsa", "south", "age"),
  bootstrap = F
)
# Can do bootstrap CIs with `bootstrap = T`
ivd$est_2sls %>% kable()
```

	Coef	SE	t	CI 2.5%	CI 97.5%	p.value
Analytic	0.0926	0.0503	1.8401	-0.006	0.1911	0.0658

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```r
ivd$F_stat
```

```
 F.standard    F.robust   F.cluster F.effective 
     9.9461      9.6316          NA      9.6316 
```

Lal et al 2024 on Weak Instruments

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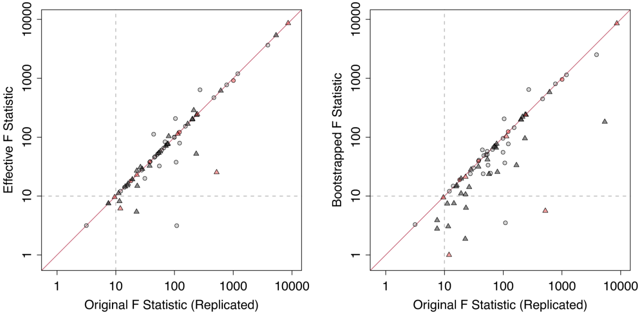

(a) Original F vs. Effective F (b) Original F vs. Bootstrapped F

Weak Instruments - tF procedure

Lee et al, 2022

- Want valid t-ratio inference for just-identified IV
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- Or equivalently, rescales the 2SLS SE by an F-dependent factor and use the standard critical value

```
ivd$tF
```

```
     F      cF    Coef      SE       t  CI2.5% CI97.5% p-value
9.6316  3.5058  0.0926  0.0503  1.8401 -0.0838  0.2689  0.3036
```

Weak Instruments - Anderson-Rubin test

Andrews et al, 2019

- Under a null $\rho = \rho_0$ is true, then after subtracting out $\rho_0 D$ from Y, the instrument Z should have no remaining predictive power for the adjusted outcome $Y - \beta_0 D$
- So we test whether Z still explains $Y - \beta_0 D$

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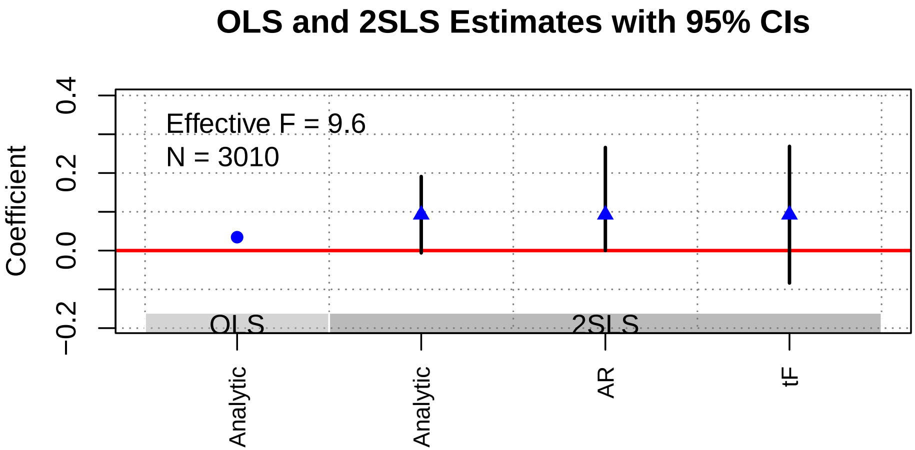
```
ivd$AR
```

```
$Fstat
        F       df1       df2         p
   3.8121    1.0000 3008.0000    0.0510

$ci.print
[1] "[0.0000, 0.2660]"

$ci
[1] 0.000 0.266

$bounded
[1] TRUE
```

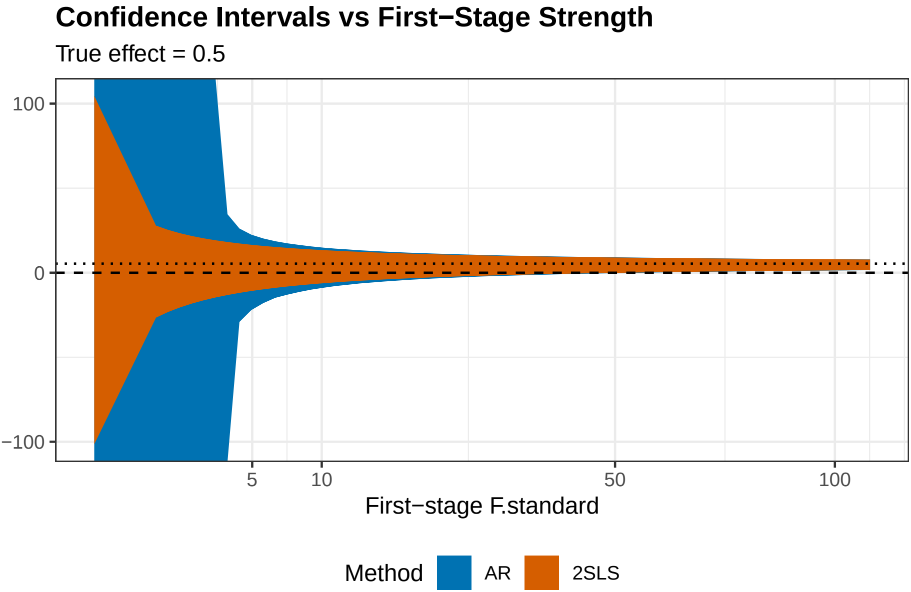
Weak Instruments

- When you have small effective F-statistic
- Use weak-instrument-robust methods (e.g., tF, Anderson–Rubin)

```
plot_coef(ivd)
```



Weak Instruments - Simulation



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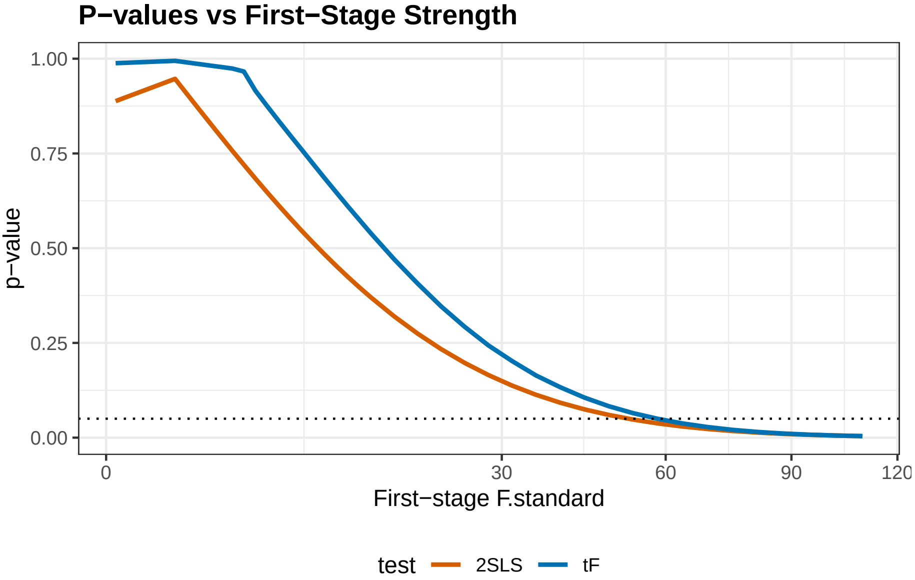

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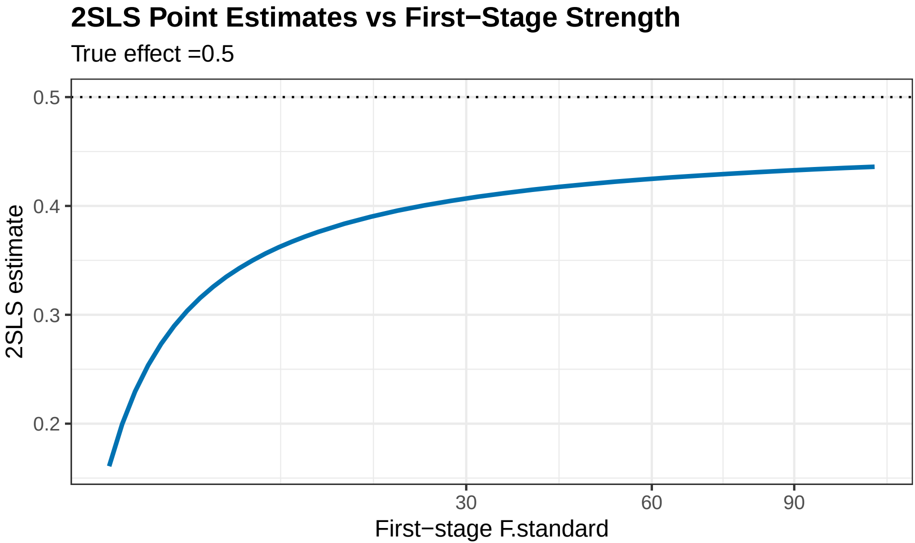

Compliance Framework - Principal Strata

4 principal strata based on treatment take-up

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- Defiers $D_i(1) = 0$, $D_i(0) = 1$

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- Never-takers: $D_i(1) = 0$, $D_i(0) = 0$
- Defiers $D_i(1) = 0$, $D_i(0) = 1$

Assume the effect of assignment on take-up is monotonic $\Rightarrow$ no defiers

LATE Theorem

Theorem: Under classic IV assumptions + no defiers:

$$\frac{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}{E[D_i|Z_i = 1] - E[D_i|Z_i = 0]} = E[Y_{1i} - Y_{0i}|D_{1i} > D_{0i}]$$

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- On the right: average causal effect of the treatment among compliers
- On the left: For binary Z, this is equivalent to IV/2SLS estimator
- LATE theorem: the IV/2SLS estimator targets the average causal effect of the treatment among compliers

Better LATE than never?

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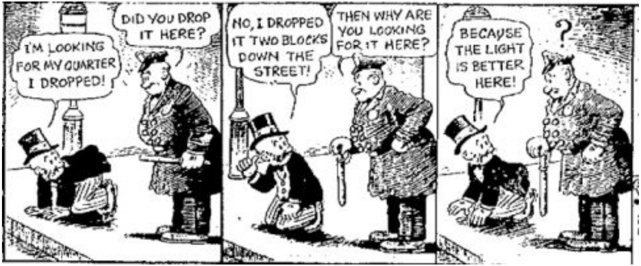
Better LATE than never?

- Angrist, Imbens, and others: We don't get the ATT or ATE but we get something that still makes some sense (particularly for policy).
- Heckman, Deaton, and others: We don't get the ATT/ATE. How does the compliers framework transfer to observational settings?



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So the "treatment effect" is $\tilde{Y}_i(D_i = 1) - \tilde{Y}_i(D_i = 0) = Y_i(1)$

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Similar logic:

- IV of $\tilde{Y}_i = Y_i(1 - D_i)$ on $\tilde{D}_i = 1 - D_i$ identifies $E[Y_0 | D_i(1) > D_i(0)]$.

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Then, compare these to baseline control outcomes $E[Y_i(0) | D_i = 0]$

Decomposing LATE

- Angrist et al 2013 shows how urban charter schools perform better than non urban charter schools

	Urban				Nonurban			
Subject	Treatment effect (1)	$E_u[Y_0 \mid D=0]$ (2)	λ_0^u (3)	λ_1^u (4)	Treatment effect (5)	$E_n[Y_0 \mid D=0]$ (6)	λ_0^n (7)	λ_1^n (8)
Panel A. Middle school								
Math	0.483***	−0.399***	0.077	0.560***	−0.177**	0.236***	0.010	−0.143***
	(0.074)	(0.011)	(0.049)	(0.054)	(0.074)	(0.007)	(0.061)	(0.042)
N	4,858				2,239			
ELA	0.188***	−0.422***	0.118**	0.306***	−0.148***	0.260***	0.102**	−0.086***
	(0.064)	(0.012)	(0.054)	(0.049)	(0.048)	(0.007)	(0.050)	(0.030)
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- Nonurban charter school students *become worse than average* at ELA and math

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Math	0.483***	−0.399***	0.077	0.560***	−0.177**	0.236***	0.010	−0.143***
	(0.074)	(0.011)	(0.049)	(0.054)	(0.074)	(0.007)	(0.061)	(0.042)
N	4,858				2,239			
ELA	0.188***	−0.422***	0.118**	0.306***	−0.148***	0.260***	0.102**	−0.086***
	(0.064)	(0.012)	(0.054)	(0.049)	(0.048)	(0.007)	(0.050)	(0.030)
N	4,551				2,323			

- $\lambda_{0/1}$ is the difference between complier un/treated potential outcomes and baseline
- Charter school students are better at ELA than average (but no difference for math) *before attending* (high λ_0)
- Nonurban charter school students *become worse than average* at ELA and math
- Urban charter school students *become better than average* at ELA and math

Characterizing Compliers

Under monotonicity

- Subjects assigned to the control group who take the treatment are "observable" always-takers

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- Implement with Abadie weights or 2SLS with constructed-outcome

Characterizing Compliers

Angrista et al 2023, Section 3.2

	Compliers			Always-takers	Never-takers
	Untreated	Treated	Pooled		
	(1)	(2)	(3)	(4)	(5)
Female	0.506	0.510	0.508	0.539	0.463
	(0.023)	(0.021)	(0.016)	(0.024)	(0.017)
Black	0.401	0.380	0.390	0.623	0.490
	(0.022)	(0.021)	(0.016)	(0.023)	(0.017)
Hispanic	0.250	0.300	0.275	0.183	0.228
	(0.02)	(0.018)	(0.013)	(0.019)	(0.014)
Asian	0.022	0.024	0.023	0.004	0.024
	(0.007)	(0.005)	(0.004)	(0.003)	(0.005)
White	0.229	0.216	0.223	0.154	0.215
	(0.018)	(0.016)	(0.012)	(0.016)	(0.014)
Special education	0.190	0.181	0.186	0.158	0.177
	(0.018)	(0.016)	(0.012)	(0.018)	(0.013)
English language learner	0.143	0.148	0.145	0.054	0.088
	(0.015)	(0.013)	(0.010)	(0.011)	(0.010)
Subsidized lunch	0.689	0.705	0.697	0.698	0.666
	(0.021)	(0.019)	(0.014)	(0.022)	(0.016)
Baseline math score	−0.274	−0.312	−0.293	−0.394	−0.301
	(0.047)	(0.041)	(0.032)	(0.045)	(0.036)
Baseline English score	−0.352	−0.349	−0.350	−0.362	−0.299
	(0.050)	(0.043)	(0.033)	(0.046)	(0.038)
Share of sample			0.546	0.197	0.257

Constructed IVs (Judge, Shift Share)

- IV doesn't have to be a directly observed variable
- Some creative designs **construct** the IV

Judge IV

- Original setting: criminal justice, effect of sentence length on future earnings
- Assignment of cases to judges quasirandom.
- Judges vary in harshness
- With this, have enough for IV. Need to use leave-one-out jacknife estimation in the first stage.
- Other settings: admissions committees, police/emergency service dispatch ...

Shift-Share IV aka Bartik Instrument

- Autor, Dorn, Hanson (2013) aka ADH

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- Unit of analysis: geographic localities ℓ

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  - Intuitively, "previous expected exposure to China imports"

Shift-Share IV Exercise

`shift-share-exercise.R`

AIPW implementation in R

Apoorva Lal 2022 Slides